RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 17 September 2026, 11:10 AM
State	Finished
Completed on	Thursday, 17 September 2026, 11:17 AM
Time taken	6 mins 45 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>


long long countWays(int n) {

    if (n == 0 || n == 1 || n == 2) {
        return 1;
    }


    long long a = 1, b = 1, c = 1;
    long long current = 0;


    for (int i = 3; i <= n; i++) {
        current = c + a;
        a = b;
        b = c;
        c = current;
    }

    return c;
}

int main() {
    int n;


    if (scanf("%d", &n) == 1) {

        printf("%lld\n", countWays(n));
    }

```

```
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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